IGCSE 0607 Practice Questions — E4.4 — Combined Questions & Markscheme

Equations of linear graphs

Total Questions: 9 — **Total Marks:** 56 Marks

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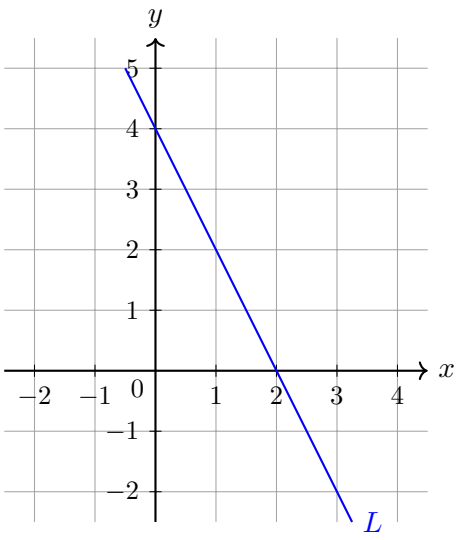
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1 Find the gradient and the y-intercept of the line with equation

$3x + 5y = 20$

Gradient = ... , y-intercept = ... **[3]**

2 The diagram shows a straight line L drawn on a coordinate grid.



(a) Find the equation of line L.
Give your answer in the form $y = mx + c$.

$y =$ ..... **[3]**

(b) Write the equation of line L in the form $ax + by = c$, where a, b and c are integers.

..... **[1]**

3 **(a)** Find the gradient and the y-intercept of the line $3x + 5y = 15$.

Gradient = ... , y-intercept = ... **[3]**

(b) Find the equation of the line parallel to $3x + 5y = 15$ that passes through the point $(0, -4)$. Give your answer in the form $y = mx + c$.

$y =$ ..... **[2]**

4 A is the point $(p, -1)$ and B is the point $(5, 2p - 1)$.
The gradient of the line AB is 3.

(a) Find the value of p.

$p =$ ..... **[3]**

(b) Find the equation of the line AB.
Give your answer in the form $y = mx + c$.

..... **[2]**

(c) The line AB intersects the x-axis at the point C and the y-axis at the point D.
Work out the area of triangle COD, where O is the origin.

..... **[3]**

5 The line L passes through the points $A(2a, a+4)$ and $B(a-1, 3a+2)$.
The gradient of line L is -4.

(a) Find the value of a.

$a =$ ..... **[3]**

(b) Find the equation of line L.
Give your answer in the form $px + qy = r$, where p, q and r are integers and $p > 0$.

..... **[3]**

(c) Line L intersects the line $y = 3x - 2$ at the point M.
Find the coordinates of M.

(..... , ...) **[3]**

6 A is the point $(-3, 1)$ and B is the point $(5, 5)$.

(a) Find the equation of the perpendicular bisector of the line segment AB.
Give your answer in the form $y = mx + c$.

$y =$ ..... **[4]**

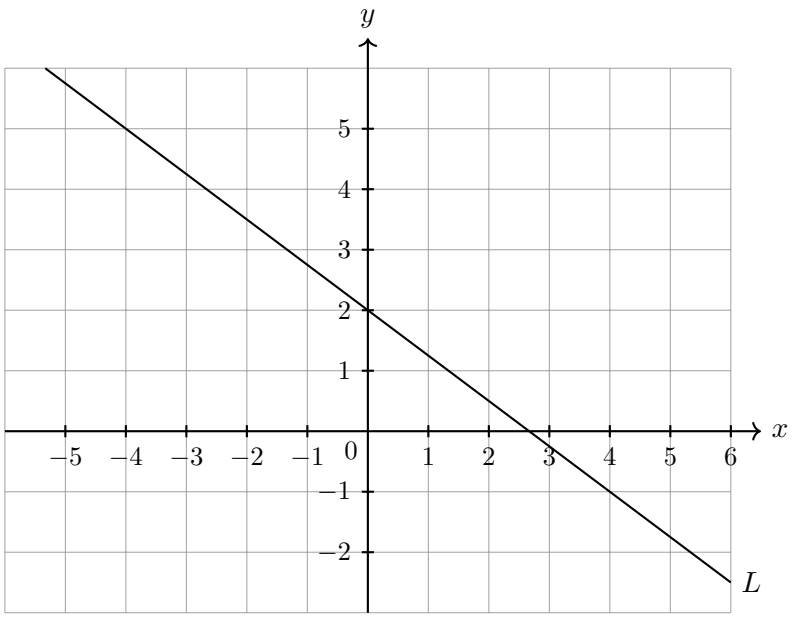
(b) The perpendicular bisector of AB intersects the line $x - 2y = 5$ at the point P.
Find the coordinates of P.

(..... , ...) **[3]**

(c) Find the exact length of AP.

..... **[3]**

7 The diagram shows a line L drawn on a coordinate grid.



(a) Work out the gradient of line L.

..... **[2]**

(b) Find the equation of line L.
Give your answer in the form $ax + by = c$, where a, b and c are integers.

..... **[3]**

8 Point A has coordinates $(-4, 3)$ and point B has coordinates $(6, -2)$.

(a) Find the equation of the line passing through A and B.
Give your answer in the form $y = mx + c$.

$y =$ ..... **[3]**

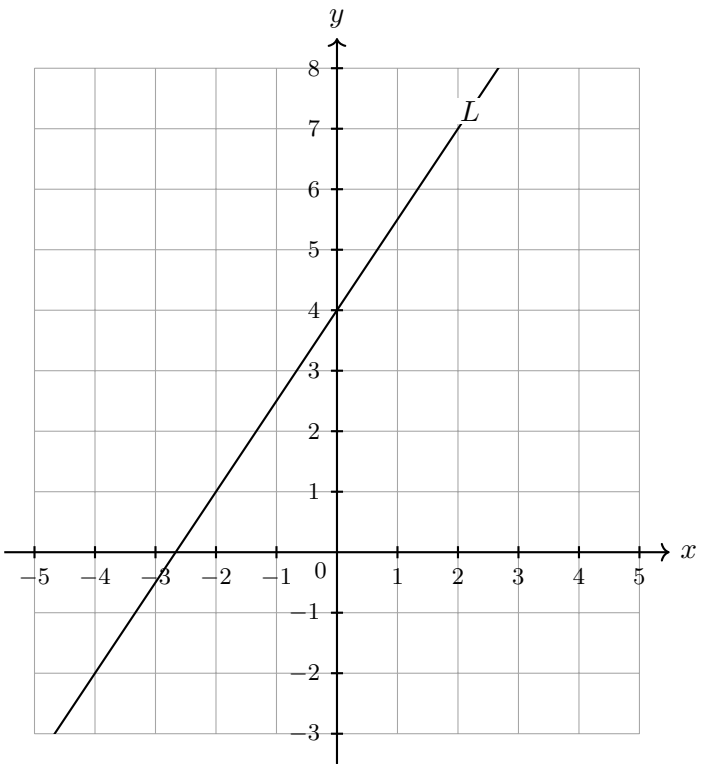
(b) The line $3x - 5y = 15$ intersects the x-axis at point C and the y-axis at point D.
Find the coordinates of C and D.

$C =$ ..... , $D =$ ..... **[2]**

(c) Write down the equation of the line that is parallel to the y-axis and passes through point A.

..... **[1]**

9 The diagram shows a straight line L.



(a) Find the equation of line L.
Give your answer in the form $ax + by = c$, where a, b, and c are integers.

..... **[3]**

(b) Line M is perpendicular to line L and passes through the point $(9, -1)$.
Find the equation of line M in the form $y = mx + c$.

$y =$ ..... **[3]**

1 Rearranging to $5y = -3x + 20$ or $y = -\frac{3}{5}x + 4$ **[M1]**

Gradient $= -\frac{3}{5}$ oe **[A1]**

y-intercept $= 4$ **[A1]**

[Back to Question] *[Total Marks: 3]*

2 **(a)** Gradient $= -2$ **[B1]**

y-intercept $= 4$ **[B1]**

$-2x + 4$ **[B1]**

(b) $2x + y = 4$ oe **[B1]**

[Back to Question] *[Total Marks: 4]*

3 **(a)** Rearranges equation to $y = mx + c$ form: $5y = -3x + 15$ or $y = -\frac{3}{5}x + 3$ **[M1]**

Gradient $= -\frac{3}{5}$ **[A1]**

y-intercept $= 3$ **[A1]**

(b) Identifies parallel gradient as $-\frac{3}{5}$ **[M1]**

$-\frac{3}{5}x - 4$ **[A1]**

[Back to Question] *[Total Marks: 5]*

4 **(a)** $\frac{(2p-1)-(-1)}{5-p} = 3$ **[M1]**

$2p = 3(5 - p)$ **[M1]**

$p = 3$ **[A1]**

(b) Substituting $p = 3$ to find coordinates $A(3, -1)$ or $B(5, 5)$ and using gradient 3 to find equation **[M1]**

$y = 3x - 10$ **[A1]**

(c) Identifies x-intercept as $\frac{10}{3}$ or y-intercept as -10 **[M1]**

$\frac{1}{2} \times \frac{10}{3} \times 10$ **[M1]**

$\frac{50}{3}$ or 16.7 $(16.66\ldots)$ **[A1]**

[Back to Question] *[Total Marks: 8]*

5 **(a)** $\frac{(3a+2)-(a+4)}{(a-1)-2a} = -4$ **[M1]**

$\frac{2a-2}{-a-1} = -4$ **[M1]**

$2a - 2 = 4a + 4$

-6 = 2a

a = -3[*A*1]

(b) Identifies coordinates of A as $(-6, 1)$ or B as $(-4, -7)$ **[M1]**

$y - 1 = -4(x + 6)$ or $y = -4x - 23$ **[M1]**

$4x + y = -23$ **[A1]**

(c) $4x + (3x - 2) = -23$ or $-4x - 23 = 3x - 2$ oe **[M1]**

$7x = -21$

x = -3[*A*1]

$y = -11$ **[A1]**

[Back to Question] *[Total Marks: 9]*

6 (a) Calculates the midpoint of $AB = \left(\frac{-3+5}{2}, \frac{1+5}{2}\right) = (1, 3)$ **[B1]**

Calculates the gradient of $AB = \frac{5-1}{5-(-3)} = \frac{4}{8} = \frac{1}{2}$ **[B1]**

Finds the perpendicular gradient $= -2$ **[M1]**

Uses perpendicular gradient -2 and midpoint $(1, 3)$ to find equation, e.g., $3 = -2(1) + c$ **[M1]**

$-2x + 5$ **[A1]**

(b) Sets up simultaneous equations, e.g., substituting $y = -2x + 5$ into $x - 2y = 5$ **[M1]**

$x - 2(-2x + 5) = 5 \implies 5x - 10 = 5 \implies x = 3$ **[M1]**

$(3, -1)$ **[A1]**

(c) Uses distance formula for $A(-3, 1)$ and $P(3, -1)$ **[M1]**

$\sqrt{(3-(-3))^2 + (-1-1)^2}$ **[M1]**

$\sqrt{40}$ or $2\sqrt{10}$ **[A1]**

[Back to Question] *[Total Marks: 10]*

7 (a) Evidence of using two integer coordinates on the line, e.g. $\frac{5-(-1)}{-4-4}$ **[M1]**

$-\frac{3}{4}$ or -0.75 **[A1]**

(b) $y = -\frac{3}{4}x + 2$ or $y - 5 = -\frac{3}{4}(x + 4)$ **[M1]**

$4y = -3x + 8$ or $4y - 20 = -3x - 12$ **[M1]**

$3x + 4y = 8$ or equivalent integer form **[A1]**

[Back to Question] *[Total Marks: 5]*

8 **(a)** Gradient $= \dfrac{-2-3}{6-(-4)}$ or $\dfrac{-5}{10} = -\dfrac{1}{2}$ **[M1]**

Substitution of $(-4, 3)$ or $(6, -2)$ into $y = -\dfrac{1}{2}x + c$, e.g. $3 = -\dfrac{1}{2}(-4) + c$ **[M1]**

$-\dfrac{1}{2}x + 1$ **[A1]**

(b) $C = (5, 0)$ **[B1]**

$D = (0, -3)$ **[B1]**

(c) $x = -4$ **[B1]**

[Back to Question] *[Total Marks: 6]*

9 **(a)** Identifies two correct points on the line, e.g., $(0, 4)$ and $(2, 7)$ **[M1]**

Calculates the gradient $m = \dfrac{7-4}{2-0} = \dfrac{3}{2}$ **[M1]**

States correct equation in required integer form, e.g., $3x - 2y = -8$ or $2y - 3x = 8$ **[A1]**

(b) Uses $m_1 \times m_2 = -1$ to find gradient of M: $-\dfrac{2}{3}$ **[M1]**

Substitutes $(9, -1)$ into $y = -\dfrac{2}{3}x + c$ to find c: $-1 = -\dfrac{2}{3}(9) + c$ **[M1]**

$-\dfrac{2}{3}x + 5$ **[A1]**

[Back to Question] *[Total Marks: 6]*